

Date: April 10, 2025

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Feature: Vanish Mode (Ephemeral Messages)

Work Completed on April 10 (3 hours):

- Implemented a **Vanish Mode toggle button** in the direct message (DM) interface with a dynamic red highlight to indicate active status.
- Integrated backend support using a `vanishMode` flag stored per DM in MongoDB.
- Modified message sending logic to flag new messages with `vanish: true` only when vanish mode is active.
- Implemented automatic **message deletion** when:
 - The recipient has seen a vanish message.
 - The recipient leaves the DM conversation.
- Used `IntersectionObserver` to track message visibility for vanish logic.
- Ensured only **the recipient** can trigger vanish deletion (not the sender).
- Created logic to **sync vanish mode state** across users by fetching it from the server on DM load and after toggling.
- Updated styling and hover effects for visual clarity using CSS.

Task: Sprint Documentation and Project Reporting

Work Completed on April 11th (2 hours):

- Compiled and maintained documentation for all features developed during the sprint, including UI components, backend endpoints, and database schema updates.
- Documented technical decisions, API designs, and user-facing functionality to ensure the project is easily understandable by stakeholders.
- Wrote detailed **system behavior descriptions** to support sprint review and project reporting.
- Contributed to the official **sprint report**, summarizing completed tasks, development progress, and challenges encountered during implementation.
- Ensured all documentation followed a consistent format and was accessible to both technical and non-technical team members.

Task: Finding 5 bugs with automatic tools

Work Completed on April 11th (2hours):

- Used **StandardJS (JavaScript Standard Style)** to automatically fix code formatting issues across the project, including indentation, unused variables, and alignment problems.

- Integrated **GitHub Advanced Security Bot (CodeQL)** to identify and fix high-severity vulnerabilities in server-side event logging.
 - Replaced template literals in console.log with safer format strings (%s) to mitigate format string injection risks.
- Addressed two critical CodeQL issues related to unescaped user input in server logs for message and userStatusUpdate events.
- Applied **GitHub Dependabot** to resolve a vulnerability related to potential middleware leaks in package-lock.json dependencies.
- Implemented and tested all tool-suggested fixes in a dedicated pull request linked to this task.